

Maitreya Patankar

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EDUCATION

San Jose State University | San Jose, CA

Aug 2025 – May 2027

Master of Science in Software Engineering, GPA: 4.0/4.0

Relevant coursework: Data Structures and Algorithms in C++(STL), Object-Oriented Programming, Operating Systems, Computer Networks, Database Systems, Data Analytics, Machine Learning, Deep Learning

TECHNICAL SKILLS

Languages: Python, Java, TypeScript, JavaScript (JS), C#, SQL, HTML/CSS

DevOps: GitHub Actions (CI/CD), Docker, Kubernetes

Cloud & Databases: AWS, Azure, PostgreSQL, MySQL, NoSQL (DynamoDB, ChromaDB)

Frameworks & Tools: React, Spring Boot, FastAPI, REST APIs, Kafka, Postman, Git

GenAI & Libraries: RAG, LangChain, LangGraph, SentenceTransformers, NumPy, scikit-learn, OpenCV

EXPERIENCE

Software Developer | BoreLeaders

Aug 2023 – Dec 2024

- Integrated shop transaction workflows into the backend system using **Python microservices**, **REST APIs**, including purchase validation and consistent server-side updates to user balances and purchased items.
- Added a new reward currency and optimized SQL queries in a **PostgreSQL** database for earning and redemption flows, reducing average API latency to under **50 ms** while keeping user balances consistent.
- Developed backend support for daily rewards and recurring weekly/biweekly live-ops events, using **DynamoDB** for flexible progress-state storage and cutting synchronization payload size by **40%**.
- Collaborated with frontend engineers from PlaySuper to integrate backend shop flows into **React** customer interfaces, ensuring seamless purchase and reward redemption user experiences across mobile and web platforms.
- Worked in **Agile/Scrum** workflow with daily standups and biweekly code reviews with a senior engineer.

Software Engineering Intern | Gameshastra

Aug 2022 – Dec 2022

- Designed a real-time multiplayer app in **C#**, implementing session flow and synchronized state updates.
- Used state synchronization for **consistency under latency**, with **turn enforcement** and **edge case handling**.
- Implemented a custom physics simulation module in **C#**, modeling forces, acceleration, and gravity for stable motion.

PROJECTS

DispatchNexus: Event-driven maintenance scheduling platform

- Decoupled ingestion from scheduling with **Kafka** to handle traffic bursts without dropping requests.
- Implemented **Spring Boot** REST endpoints for job scheduling and fulfillment and added **JUnit** tests to validate request/response behavior, improving **p95** latency of scheduling API by **30%** and reducing regression risk.
- Automated build, test and deploy with **GitHub Actions** and added **Prometheus/Grafana** dashboards to improve service observability and speed up issue triage
- Engineered a **LangGraph** stateful **multi-agent workflow** that negotiated with providers and retried assignments until acceptance, fully automating scheduling.

SAM: Apartment management platform

- Implemented **RAG** over 35+ policy documents using **ChromaDB**, achieving retrieval latency under **100 ms**. Tracked vector search and end-to-end request latency in **CloudWatch** to catch regressions.
- Built **FastAPI** REST services with a **React/TypeScript** and **Vue.js** frontend and added **pytest** coverage for key API paths to validate retrieval and response behavior, improving reliability during iteration.
- Used **AWS CloudWatch** (metrics, logs, dashboards, alarms) to monitor p50/p95 latency, error rate, throughput, and service health checks, and analyzed failure patterns to debug issues and improve reliability.
- Trained an **XGBoost** model to classify resident messages into predefined categories using **NLP**, providing structured inputs that improved LLM response selection and reliability.